

Week 9 – House Price Prediction

1. Project Overview

This project demonstrates the fundamentals of Machine Learning by building a House Price Prediction model.

The objective is to predict house prices using features such as area, bedrooms, bathrooms, age, location and property type.

2. Objectives

- Load and inspect the house price dataset.
- Clean and prepare the data.
- Handle categorical features.
- Split the data into training and testing sets.
- Implement Linear Regression.
- Implement Linear Regression from scratch.
- Evaluate model performance.
- Compare actual and predicted prices.
- Identify important features affecting house prices.

3. Dataset

Dataset:

`data/house\_prices.csv`

The dataset contains house-related information including:

- Property ID
- Area
- Bedrooms
- Bathrooms
- Age
- Location
- Property Type
- Price

The target variable is **Price**.

4. Technologies Used

- Python
- Pandas
- NumPy

- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

5. Data Preprocessing

The dataset was inspected for:

- Missing values
- Duplicate records
- Data types
- Numerical and categorical columns
- Basic statistical information

Categorical features such as Location and Property Type were encoded before training the model.

The dataset was then divided into training and testing sets.

6. Machine Learning Model

Linear Regression

Linear Regression was used to predict house prices based on the selected features.

Two approaches were demonstrated:

1. Linear Regression implemented from scratch.
2. Linear Regression using Scikit-learn.

The Scikit-learn implementation was used for the final model evaluation.

7. Model Evaluation

The model was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R² Score

The model predictions were compared with the actual house prices to understand prediction performance.

The exact evaluation values are generated automatically by the project during execution.

8. Visualization

The project includes visualizations for understanding the model and dataset.

Generated visualizations include:

- Area vs Price
- Actual vs Predicted Prices

- Feature Analysis

All visualizations are stored inside:

\visualizations\

```
Task_9_Machine_Learning_Concepts > house_price_prediction.py > ...
23 def load_and_validate_data() -> pd.DataFrame:
24     """Load the real dataset and validate its required structure."""
25     if not DATA_FILE.exists():

D:\Developer_Arena\Week1_Task\Task_9_Machine_Learning_Concepts>py house_price_prediction.py
House price prediction completed successfully.
Dataset shape: (300, 8)
Training samples: 240
Testing samples: 60

Scikit-learn Linear Regression metrics:
MAE: 2,188,736.34
MSE: 8,454,330,868,276.57
R2 Score: 0.9406

Scratch Linear Regression metrics:
MAE: 2,188,736.34
MSE: 8,454,330,868,457.58
R2 Score: 0.9406

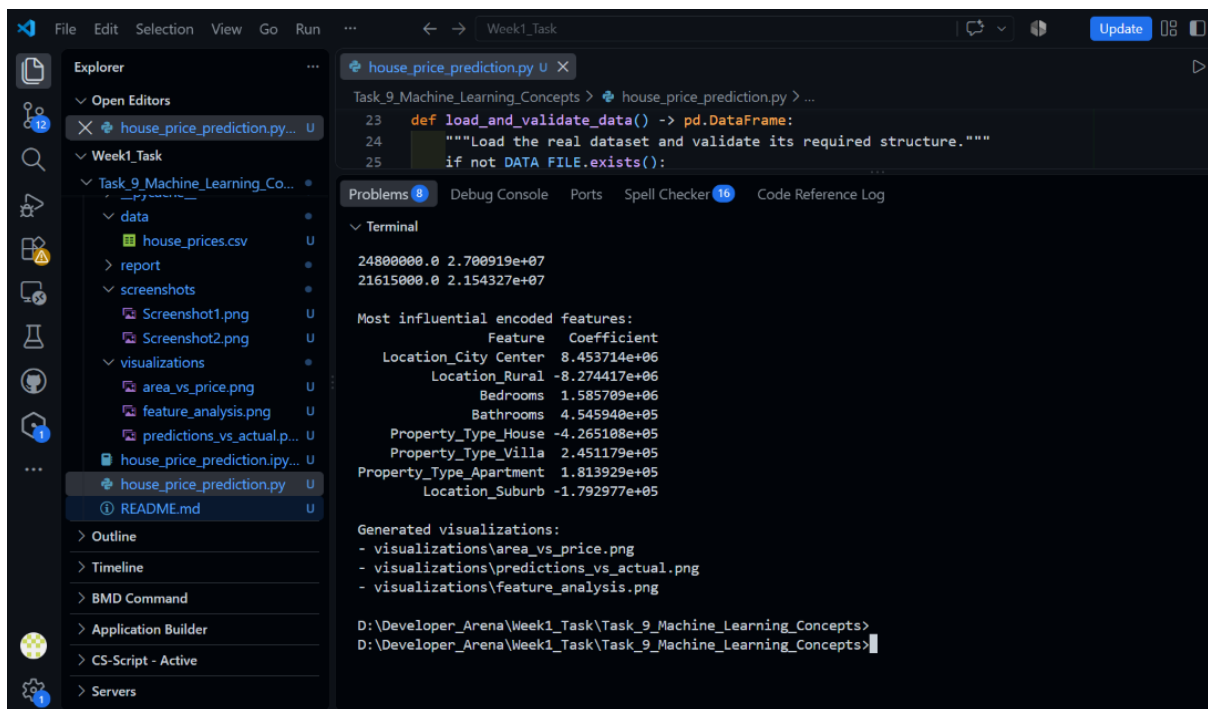
Actual vs predicted examples:
Actual    Predicted
8930000.0  1.507481e+07
18085000.0 1.916367e+07
```

```
Task_9_Machine_Learning_Concepts > house_price_prediction.py > ...
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Actual vs predicted examples:
Actual    Predicted
8930000.0  1.507481e+07
18085000.0 1.916367e+07
53980000.0 4.932125e+07
24800000.0 2.700919e+07
21615000.0 2.154327e+07

Most influential encoded features:
Feature          Coefficient
Location_City_Center  8.453714e+06
Location_Rural      -8.274417e+06
Bedrooms            1.585709e+06
Bathrooms           4.545940e+05
Property_Type_House -4.265108e+05
Property_Type_Villa  2.451179e+05
Property_Type_Apartment 1.813929e+05
Location_Suburb     -1.792977e+05

Generated visualizations:
- visualizations\area_vs_price.png
```



9. Feature Analysis

The trained model coefficients were analyzed to understand the contribution of different features to house prices.

Features such as area, number of bedrooms, bathrooms, age, location and property type were considered during the analysis.

10. Project Structure

Task_9_Machine_Learning_Concepts/

```
|
|— data/
|   |— house_prices.csv
|
|— visualizations/
|   |— area_vs_price.png
|   |— predictions_vs_actual.png
|   |— feature_analysis.png
|
|— report/
|   |— model_evaluation_report.md
|
|— screenshots/
|
|— house_price_prediction.ipynb
```

|— house_price_prediction.py

|— README.md

|— requirements.txt